Fish oil supplementation in type 2 diabetes: a quantitative systematic review.

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OBJECTIVE: To determine the effects of fish oil supplementation on lipid levels and glycemic control in patients with type 2 diabetes. RESEARCH DESIGN AND METHODS: A comprehensive search of Medline, Embase, Lilacs, the Cochrane Clinical Trials Registry bibliographies of relevant papers, and expert input updated through September 1998 was undertaken. All randomized placebo-controlled trials were included in which fish oil supplementation was the only intervention in subjects with type 2 diabetes. Three investigators performed data extraction and quality scoring independently with discrepancies resolved by consensus. Eighteen trials including 823 subjects followed for a mean of 12 weeks were included. Doses of fish oil used ranged from 3 to 18 g/day The outcomes studied were glycemic control and lipid levels. RESULTS: Meta-analysis of pooled data demonstrated a statistically significant effect of fish oil on lowering triglycerides (-0.56 mmol/l [95% CI -0.71 to -0.41]) and raising LDL cholesterol (0.21 mmol/l [0.02 to 0.41]). No statistically significant effect was observed for fasting glucose. HbA1c total cholesterol, or HDL cholesterol. The triglyceride-lowering effect and the elevation in LDL cholesterol were most marked in those trials that recruited hypertriglyceridemic subjects and used higher doses of fish oil. Heterogeneity was observed and explained by the recruitment of subjects with baseline hypertriglyceridemia in some studies. CONCLUSIONS: Fish oil supplementation in type 2 diabetes lowers triglycerides, raises LDL cholesterol, and has no statistically significant effect on glycemic control. Trials with hard clinical end points are needed.